- Understand how to formulate linear-integer optimization models.
- Understand how to solve linear-integer optimization models, both by hand and with the computer through an algebraic modeling language and a state-of-the-art solver.
- Understand the special structure underlying linear-integer optimization models and how this affects their ability to be solved.
- Understand decomposition techniques to aid in solution.

ORWE598. SPECIAL TOPICS. 0-6 Semester Hr.

ORWE598. ALGORITHMS FOR OPERATIONS RESEARCH. 0-6 Semester Hr.
Reasoning about algorithm correctness (proofs, counterexamples). Analysis of algorithms: asymptotic and practical complexity. Review of dictionary data structures (including balanced search trees). Priority queues. Advanced sorting algorithms (heapsort, radix sort). Advanced algorithmic concepts illustrated through sorting (randomized algorithms, lower bounds, divide and conquer). Dynamic programming. Backtracking. Algorithms on unweighted graphs (traversals) and weighted graphs (minimum spanning trees, shortest paths, network flows and bipartite matching); NP-completeness and its consequences. Prerequisite: CSCI220 with a grade of C- or higher or CSCI262 with a grade of C- or higher, MATH213 or MATH223 or MATH224, MATH300 or MATH358 or CSCI358.
Course Learning Outcomes

- Same as existing CSCI406.

ORWE598. SIMULATION FOR OPERATIONS RESEARCH. 0-6 Semester Hr.
A first course in computer simulation using formal learning groups and emphasizing the rigorous development of simulation applications. Topics will include random number generation, Monte Carlo simulation, discrete event simulation, and the mathematics behind their proper implementation and analysis (random variates, arrival time modeling, infinite horizon statistics, batch means and sampling techniques). The course uses learning group assignments, quizzes, programming projects (using Linux) and exams for assessment. Prerequisite: (CSCI210 or CSCI274) AND CSCI306 AND (MATH201 or MATH334).
Course Learning Outcomes

- same as existing course CSCI423

ORWE599. INDEPENDENT STUDY. 0.5-6 Semester Hr.

ORWE686. ADVANCED LINEAR OPTIMIZATION. 3.0 Semester Hrs.
(II) As an advanced course in optimization, we expand upon topics in linear programming: advanced formulation, the dual simplex method, the interior point method, algorithmic tuning for linear programs (including numerical stability considerations), column generation, and Dantzig-Wolfe decomposition. Time permitting, dynamic programming is introduced. Applications of state-of-the-art hardware and software emphasize solving real-world problems in areas such as manufacturing, mining, energy, transportation and logistics, and the military. Computers are used for model formulation and solution. Offered every other year. Prerequisite: MEGN586. 3 hours lecture; 3 semester hours.
Course Learning Outcomes

- Understand how to formulate complicated linear optimization models.
- Dual Simplex Method and Interior Point Method
- Algorithmic Tuning
- Column Generation and Dantzig-Wolfe Decomposition

ORWE688. ADVANCED INTEGER OPTIMIZATION. 3.0 Semester Hrs.
(II) As an advanced course in optimization, we expand upon topics in integer programming: advanced formulation, strong integer programming formulations (e.g., symmetry elimination, variable elimination, persistence), in-depth mixed integer programming cuts, rounding heuristics, constraint programming, and decompositions. Applications of state-of-the-art hardware and software emphasize solving real-world problems in areas such as manufacturing, mining, energy, transportation and logistics, and the military. Computers are used for model formulation and solution. Prerequisite: MEGN588. 3 hours lecture; 3 semester hours. Offered every other year.
Course Learning Outcomes

- Know how to formulate advanced integer optimization models 2. Be familiar with advanced algorithms to solve these models 3. Be able to use software, including scripting, to model and solve these models 4. Understand the theory behind and mathematical tenants of advanced integer optimization models

Program Director

Alexandra Newman, Professor, Mechanical Engineering

Quantitative Biosciences and Engineering

Degrees Offered

- Master of Science in Quantitative Biosciences and Engineering (non-thesis)
- Master of Science in Quantitative Biosciences and Engineering (thesis)
- Doctor of Philosophy in Quantitative Biosciences and Engineering

Program Description

The graduate program in quantitative biosciences and engineering brings together faculty across the Mines campus working on diverse areas of biology to educate students with at least a Bachelor of Science degree in engineering or science in the diverse field of biology. Biology deals broadly with life on this planet, the human organism and its health, and harnessing biological processes to produce fuels, chemicals, and consumer products. Thus, biology in general and human health and well-being in particular are important application areas for virtually all other areas of science, technology, and engineering. This is reflected in the fact that any academic discipline exists today with a bio prefix, such as biophysics, biochemistry, bioengineering, mathematical biology, computational biology, systems biology, structural biology, biomedicine, biomaterials, biomechanics, bioinformatics, biological chemistry, geobiology, environmental biology, microbiology, to name just a few. Similarly, health is included in many labels, e.g., digital healthcare, health economics, and health informatics. Educating students at the interfaces of biology, health, and engineering with other disciplines is a primary goal of this program.

Many departments at Mines jointly administer this cross-departmental program in quantitative biosciences and engineering. The program coexists alongside strong disciplinary programs, in chemistry and geochemistry, chemical and biochemical engineering, physics, computer science, mathematics and statistics, mechanical engineering and metallurgical and materials engineering, civil and environmental engineering, economics, geology, and geological engineering and geophysics, and thus draws from the strengths of these programs

through close links and joint courses. For administrative purposes, at the graduate level, the student will reside in the advisor's home academic department. The student's graduate committee will have final approval of the course of study.

Fields of Research

Research at Mines in this rapidly growing field currently includes but is not limited to the following general areas:

- Laser Design and Imaging
- Biofuels and Metabolic Engineering
- Omics and Systems Biology
- Environmental Toxicology and Microbiology
- Biosensors and Devices
- Biotechnology
- Biomechanics
- Biofluid mechanics
- Bioinformatics and Computational Biology
- Tissue Engineering & Biomaterials
- Physical Biochemistry
- Biophysics and Analytical Methodology Development
- Digital Healthcare
- Mathematical Biology

More than 35 faculty members across the Mines campus participate in this program, which will in the future also involve faculty of nearby collaborating institutions and scientists from the biotech/healthcare industry.

Quantitative Biosciences and Engineering (QBE) Program Requirements

For admission, students may enter with biology or health-related undergraduate degrees or with a technical degree, e.g., in engineering, mathematics, or computer science.

Current Mines undergraduate students have the option to apply to the Office of Graduate Studies for the combined program while pursuing their undergraduate degree (see information below).

Each of the three degrees (non-thesis Master of Science, thesis-based Master of Science, and Doctor of Philosophy) require the successful completion of four core courses for a total of 13 credits, as detailed below.

QBE Core Courses		
BIOL500	CELL BIOLOGY AND BIOCHEMISTRY	4.0
BIOL510	BIOINFORMATICS	3.0
BIOL520	SYSTEMS BIOLOGY	3.0
CHGN535	PHYSICAL BIOCHEMISTRY	3.0
Total Semester Hrs		**13.0**

QBE Graduate Seminar

Full-time graduate students in the QBE program are expected to maintain continuous enrollment in BIOL 590, QBE Graduate Seminar, a 1 credit course. A maximum of 2 credits will be granted toward the MS degree requirements while a maximum of 4 credits will be granted toward PhD requirements, as shown below. Students who are concurrently enrolled in a different degree program that also requires seminar attendance may have this requirement waived at the discretion of the QBE Program Director.

Master of Science in Quantitative Biosciences and Engineering (Non-Thesis Option)

The Master of Science Non-Thesis (MS-NT) degree requires a minimum of 30 credits of acceptable coursework.

QBE Core Courses		13.0
QBE Electives (see list below)		15.0
BIOL590	QUANTITATIVE BIOSCIENCES & ENGINEERING GRADUATE SEMINAR (*)	2.0
Total Semester Hrs		**30.0**

*While full-time MS-NT students are expected to maintain continuous enrollment in BIOL 590, the QBE Graduate Seminar; a maximum of 2 credits will be granted toward the MS-NT degree requirements.

Master of Science in Quantitative Biosciences and Engineering (Thesis Option)

The thesis-based Master of Science (MS-T) requires a minimum of 30 semester hours of acceptable coursework and thesis research credits. Students conduct an in-depth research project with one of the participating faculty members who are currently accepting masters degree students. The student must also submit a thesis and pass the thesis defense examination before the thesis committee.

QBE Core Courses		13.0
QBE Elective		3.0
BIOL590	QUANTITATIVE BIOSCIENCES & ENGINEERING GRADUATE SEMINAR (*)	2.0
BIOL707	GRADUATE THESIS / DISSERTATION RESEARCH CREDIT	12.0
Total Semester Hrs		**30.0**

*While full-time MS-T students are expected to maintain continuous enrollment in BIOL 590, the QBE Graduate Seminar; a maximum of 2 credits will be granted toward the MS-T degree requirements.

Doctor of Philosophy in Quantitative Biosciences and Engineering

The Doctor of Philosophy (PhD)degree requires a minimum of 72 hours of course and research credit including at least 24 credits in coursework and at least 24 credits in research. Doctoral students must also pass a qualifying examination and thesis-proposal defense, complete a satisfactory thesis, and successfully defend their thesis.

QBE Core Courses	13.0
QBE Electives	11.0

BIOL590	QUANTITATIVE BIOSCIENCES & ENGINEERING GRADUATE SEMINAR (*)	4.0
BIOL707	GRADUATE THESIS / DISSERTATION RESEARCH CREDIT	24.0
QBE Electives or BIOL707 Research		20.0
Total Semester Hrs		**72.0**

*While full-time PhD students are expected to maintain continuous enrollment in BIOL 590, the QBE Graduate Seminar, a maximum of 4 credits will be granted toward the PhD degree requirements.

QBE Elective Courses:

The current list of available electives is shown below. Because course options are continually expanding, additional complementary courses (beyond those listed here) may be approved on an ad hoc basis by the advisor in consultation with the program director.

BIOL599	INDEPENDENT STUDY	0.5-6
CBEN505	NUMERICAL METHODS IN CHEMICAL ENGINEERING	3.0
CBEN511	NEUROSCIENCE, MEMORY, AND LEARNING	3.0
CBEN531	IMMUNOLOGY FOR SCIENTISTS AND ENGINEERS	3.0
CBEN532	TRANSPORT PHENOMENA IN BIOLOGICAL SYSTEMS	3.0
CBEN570	INTRODUCTION TO MICROFLUIDICS	3.0
CBEN624	APPLIED STATISTICAL MECHANICS	3.0
CBEN625	MOLECULAR SIMULATION	3.0
CBEN670	ADVANCED MICROSCOPY FOR RESEARCH	
CEEN501	LIFE CYCLE ASSESSMENT	3.0
CEEN550	PRINCIPLES OF ENVIRONMENTAL CHEMISTRY	3.0
CEEN551	ENVIRONMENTAL ORGANIC CHEMISTRY	3.0
CEEN560	MOLECULAR MICROBIAL ECOLOGY AND THE ENVIRONMENT	3.0
CEEN562	ENVIRONMENTAL GEOMICROBIOLOGY	3.0
CEEN566	MICROBIAL PROCESSES, ANALYSIS AND MODELING	3.0
CEEN570	WATER AND WASTEWATER TREATMENT	3.0
CHGN509	BIOLOGICAL INORGANIC CHEMISTRY	3.0
CHGN507	ADVANCED ANALYTICAL CHEMISTRY	3.0
CSCI562	APPLIED ALGORITHMS AND DATA STRUCTURES	3.0
CSCI/DSCI575	ADVANCED MACHINE LEARNING	3.0
DSCI503	ADVANCED DATA SCIENCE	3.0
DSCI570	INTRODUCTION TO MACHINE LEARNING	
EBGN525	BUSINESS ANALYTICS	3.0
EBGN553	PROJECT MANAGEMENT	3.0
MATH530	INTRODUCTION TO STATISTICAL METHODS	3.0
MATH/DSCI560	INTRODUCTION TO KEY STATISTICAL LEARNING METHODS I	
MATH572	MATHEMATICAL AND COMPUTATIONAL NEUROSCIENCE	3.0
MEGN532	EXPERIMENTAL METHODS IN BIOMECHANICS	3.0
MEGN535	MODELING AND SIMULATION OF HUMAN MOVEMENT	3.0
MEGN536	COMPUTATIONAL BIOMECHANICS	3.0
MTGN570	BIOCOMPATIBILITY OF MATERIALS	3.0
MTGN572	BIOMATERIALS	3.0

Mines' Combined Undergraduate / Graduate Degree Program

Students enrolled in Mines' combined undergraduate/graduate program may double count up to 6 credits of graduate coursework to fulfill requirements of both their undergraduate and graduate degree programs. These courses must have been passed with B- or better, not be substitutes for required coursework, and meet all other university, department, and program requirements for graduate credit.

Students are advised to consult with their undergraduate and graduate advisors for appropriate courses to double count upon admission to the combined program.

BIOL500. CELL BIOLOGY AND BIOCHEMISTRY. 4.0 Semester Hrs.
This graduate-level course offers an in-depth exploration of the biochemical processes governing cellular function, structure, and metabolism. Bridging molecular structure with system-level function, students will examine critical topics including cell components, membrane dynamics, gene expression, signal transduction and metabolic pathways. Through rigorous literature analysis and reviewing biochemistry, cell and molecular biology techniques, this course prepares students for future research careers, exploring complex mechanisms behind apoptosis and cancer biology. The course includes a 3-credit hour lecture section and a 1 credit hour practicum section.
Course Learning Outcomes

- Relate 3D structure of biological macromolecules to their functional roles in cellular processes.
- Describe the molecular mechanisms governing eukaryotic gene expression, including transcription, RNA processing, and translation.
- Describe major metabolic pathways and how they are regulated under different physiological conditions and in diseases.
- Analyze the mechanisms of intracellular communication and signaling cascades that regulate cell growth, survival, and death.
- Describe the molecular and genetic basis of tumorigenesis, including the role of oncogenes, tumor suppressors, and altered metabolic states in cancer cells.
- Integrate knowledge of molecular mechanisms to understand complex, system-level functions, such as cell-to-cell communication and tissue-level responses.
- Critically analyze and interpret primary literature in biochemistry, cell and molecular biology to evaluate scientific findings.
- Review and explain the theory behind modern research techniques used to study cell components, such as mass spectrometry, imaging, and molecular cloning.
- Develop original research questions and propose appropriate experimental approaches to test them.
- Prepare for future research careers by understanding the methodologies used in current cancer biology and cell biology research.
- Present complex scientific information, techniques, and experimental findings in various formats.

BIOL501. ADVANCED BIOCHEMISTRY. 3.0 Semester Hrs.
Advanced study of the major molecules of biochemistry: amino acids, proteins, enzymes, nucleic acids, lipids, and saccharides- their structure,

chemistry, biological function, biosynthesis, and interaction. Stresses bioenergetics and the cell as a biological unit of organization. Advanced discussion of the intertwining of molecular genetics, biomolecule synthesis, and metabolic cycles. Prerequisites: CHGN428 or BIOL500.
Course Learning Outcomes

- Demonstrate a broad knowledge of the fundamental introductory concepts of Chemistry, Biology and Physics
- Demonstrate a thorough knowledge of the intersection between the disciplines of Biology and Chemistry.
- Locate, critically analyze, interpret and discuss data, hypotheses, results, theories, and explanations found in the primary literature, applying knowledge from Chemistry and Biology.
- Appreciate the way in which practitioners in the disciplines of Biology and Chemistry intersect and bring their expertise to bear in solving complex problems involving living systems.

BIOL510. BIOINFORMATICS. 3.0 Semester Hrs.
Bioinformatics is a blend of multiple areas of study including biology, data science, mathematics and computer science. The field focuses on extracting new information from massive quantities of biological data and requires that scientists know the tools and methods for capturing, processing and analyzing large data sets. Bioinformatics scientists are tasked with performing high-throughput, next-generation sequencing. They analyze DNA sequence alignment to find mutations and anomalies and understand the impact on cellular processes. The bioinformatician uses software to analyze protein structure and its impact on cell function. Learning how to design experiments and perform advanced statistical analysis is essential for anyone interested in this field, which is main goal of this course. Prerequisite: CSCI102.
Course Learning Outcomes

- 1. knowledge and awareness of the basic principles and concepts of biology, computer science and mathematics;
- 2. existing software effectively to extract information from large databases and to use this information in computer modeling;
- 3. problem-solving skills, including the ability to develop new algorithms and analysis methods;
- 4. an understanding of the intersection of life and information sciences, the core of shared concepts, language and skills;
- 5. the ability to speak the language of structure-function relationships, information theory, gene expression, and database queries.

BIOL520. SYSTEMS BIOLOGY. 3.0 Semester Hrs.
This course provides students an introduction to the emerging field of systems biology. It will consist of lectures, group discussion sessions, and problem-solving sessions and/or computational labs. Students will learn strategies and tools to interrogate biological systems using mathematical modeling. Topics of the course will come from typical aspects of biomathematical modeling including, but not limited to: the choice of a modeling framework from various approaches; the design of interaction diagrams; the identification of variables and processes; the design of systems models; standard methods of parameter estimation; the analysis of steady states, stability, sensitivity; numerical evaluations of transients; phase-plane analysis; simulation of representative biological scenarios. All theoretical concepts are exemplified with applications.
Course Learning Outcomes

- At the completion of the course, students will be able to:1. Describe and understand important types of quantitative/mathematical models used in the field of systems biology
- 2. Explain the basic strengths and limitations of quantitative/mathematical modeling of biological systems
- 3. Design and implement quantitative/mathematical models of biological systems
- 4. Apply appropriate techniques for steady-state and dynamical analysis of models
- 5. Utilize different modeling tools for the analysis of models and their output
- 6. Assimilate current systems biology literature, extend it in a final project, and communicate results professionally and effectively

BIOL590. QUANTITATIVE BIOSCIENCES + ENGINEERING GRADUATE SEMINAR. 1.0 Semester Hr.
(I,II) The Quantitative Biosciences and Engineering (QBE) Graduate Seminar provides a forum for QBE graduate students to participate in seminars given by QBE professionals, develop an enhanced understanding of the breadth of quantitative bioscience disciplines, and present their research projects. Grade is based on attendance over the semester. Full-time graduate students must enroll in the Graduate Seminar course every semester that they are enrolled at Mines. Repeatable; maximum 2 credits granted towards MS degree requirements and 4 credits maximum granted towards PhD Requirements.

BIOL707. GRADUATE THESIS / DISSERTATION RESEARCH CREDIT. 1-15 Semester Hr.
(I, II, S) Research credit hours required for completion of a Masters-level thesis or Doctoral dissertation. Research must be carried out under the direct supervision of the student's faculty advisor. Variable class and semester hours. Repeatable for credit.

Advising Faculty

Joel Bach, Emeritus Associate Professor of Mechanical Engineering

Parisa Bazazi, Assistant Professor of Petroleum Engineering

Cecilia Diniz Behn, Professor of Applied Mathematics & Statistics

Nanette Boyle, Department Head and Professor of Chemical and Biological Engineering

Kevin Cash, Director of the QBE Graduate Program and Associate Professor of Chemical and Biological Engineering

Anuj Chauhan, Professor of Chemical and Biological Engineering

Dylan Domaille, Associate Professor of Chemistry

Linda Figueroa, Professor of Civil and Environmental Engineering

Christopher Higgins, Professor of Civil and Environmental Engineering

Katie Knaus , Assistant Professor of Mechanical Engineering

Melissa Krebs, Associate Professor of Chemical and Biological Engineering

Ramya Kumar, Assistant Professor of Chemical and Biological Engineering

Terry Lowe, Research Professor of Materials and Metallurgical Engineering

David Marr, Professor of Chemical and Biological Engineering